# Statistics
# Parameter Estimation, Hypothesis Testing, Regression

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# Lecture 3

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# Esimation

- Debugging a program: A program contains an unknown number of bugs, say $N$.
- Deliberately introduce $n$ (known) bugs and see how many of these are caught, say $n_1$. Estimate $N$.
- Equivalent to calculating the volume of an arbitrary object $N$-dimensions with a membership oracle and a randomised algorithm.
- Other examples from before: Fish population in the Lake. $n$ fish in the lake are marked. $n$ are caught after some time and the number of marked fish is counted, say $n_1$. The experiment (catching the fish) is repeated $K$ times with sample values $n_1, \ldots, n_K$. Estimate the number of fish in the Lake.

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# Estimators: Point and Interval Estimate

- Consider a population. Interest is to estimate a parameter associated with the population.
- First step is to obtain $n$ **random** samples from the population and measure the parameter of interest.
- Let $\mathsf{X}_i$ be the random variable denoting the possible value from the $i$-th sample.
- Let $\theta$ be the parameter of interest.
- Let $x = \{x_1, x_2, \ldots, x_n\}$ be values from the sample measurements
- $x_i$ is *realisation* of random variable $\mathsf{X}_i$.
- Any function of the samples $x$ is called a *statistic*.

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# Estimators: Point and Interval Estimate

- $\hat{\theta} = g(x_1, x_2, \ldots, x_n)$ is a *point estimate* of $\theta$.
- $\left(\mathsf{E}\left(\hat{\theta}\right) - \theta\right)$ is called the **bias** of the estimate.
- $\mathsf{E}\left(\left(\hat{\theta} - \theta\right)^2\right)$ is called the mean squared error of the estimate.
- An alterntaive to a point estimate is the *interval estimate* described using $\theta_l, \theta_u, \gamma$

$$\mathsf{Prob}(\theta_l \leq \theta \leq \theta_u) \geq \gamma$$

- $\gamma$ is called the *confidence level* and $(\theta_l, \theta_u)$ is the the *interval estimate.*

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# Recap: Sums of Independent Random Variables

- Consider the sum of $n$ independent and identically distributed (iid) random variables—$\mathsf{X}_i$, $i = 1, \ldots n$.
- Let $\mathsf{E}(\mathsf{X}_1) = \mu$ and $\mathsf{VAR}(\mathsf{X}_1) = \sigma^2$.
- Let the sample sum be $\mathsf{S}_n := \sum_{i=1}^{n} \mathsf{X}_i$.
- We know that $\mathsf{E}(\mathsf{S}_n) = n\mu$ and $\mathsf{VAR}(\mathsf{S}_n) = n\sigma^2$.
- From the Law of Large Numbers, $\frac{\mathsf{S}_n}{n} \longrightarrow \mu$,
  - Intuitively, if we take a large number of samples, the sample mean will 'approach the true mean.

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# Recap: Sums of Independent Random Variables

- From Central Limit Theorem, with Z being the unit Normal,

$$\text{Prob}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} < y\right) \rightarrow \text{Prob}(Z < y)$$

  - Crudely, as $n$ becomes large, the distribution of the sum approaches that of the Gaussian distribution with same mean and same variance as the sum.
  - Thus for large $n$, the Gaussian is a good approximation for the distribution of the sum of $n$ iid random variables.

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# The Unit Normal

- $\mathsf{Z}$ is the zero-mean Gaussian random variable with unit variance.
- If $\mathsf{X}$ is a Gaussian random variable with mean $\mu$ and standard deviation $\sigma$, then $\frac{\mathsf{X}-\mu}{\sigma}$ is unit normal.
- The above operation essentially *shifts and scales* the original Gaussian, it normalises it.
- Some 'unit normal' probabilities
  - $\mathsf{Prob}(|\mathsf{Z}| < 1) \approx 0.68$
  - $\mathsf{Prob}(|\mathsf{Z}| < 2) \approx 0.95$
  - $\mathsf{Prob}(|\mathsf{Z}| < 3) \approx 0.997$
  - $\mathsf{Prob}(|\mathsf{Z}| > 1) \approx 1 - 0.68 = 0.32$

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# The Unit Normal

- Prob$(0 \leq \mathsf{Z} < 1) \approx ?$
- Prob$(\mathsf{Z} \notin (0, 1)) \approx ?$
- Prob$(\mathsf{Z} \notin (0, 2)) \approx ?$
- Prob$(\mathsf{Z} > z_\alpha) = \alpha$

| Percentile | 90 | 95 | 97.5 | 99 | 99.5 | 99.9 | 99.95 |
|---|---|---|---|---|---|---|---|
| $\alpha$ | 0.1 | 0.05 | 0.025 | 0.01 | 0.005 | 0.001 | 0.0005 |
| $z_\alpha$ | 1.28 | 1.645 | 1.96 | 2.33 | 2.58 | 3.08 | 3.27 |

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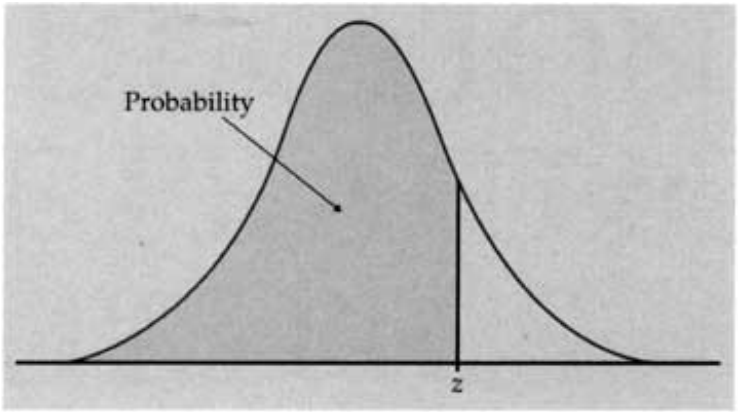



**Table entry for $z$ is the area under the standard normal curve to the left of $z$.**

TABLE A Standard normal probabilities (*continued*)

| $z$ | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| 0.1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| 0.2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| 0.3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| 0.4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| 0.5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| 0.6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| 0.7 | .7580 | .7611 | .7642 | .7673 | .7704 | .7734 | .7764 | .7794 | .7823 | .7852 |
| 0.8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8078 | .8106 | .8133 |
| 0.9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9236 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9484 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9761 | .9767 |
| 2.0 | .9772 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |
| 2.1 | .9821 | .9826 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9854 | .9857 |
| 2.2 | .9861 | .9864 | .9868 | .9871 | .9875 | .9878 | .9881 | .9884 | .9887 | .9890 |
| 2.3 | .9893 | .9896 | .9898 | .9901 | .9904 | .9906 | .9909 | .9911 | .9913 | .9916 |
| 2.4 | .9918 | .9920 | .9922 | .9925 | .9927 | .9929 | .9931 | .9932 | .9934 | .9936 |
| 2.5 | .9938 | .9940 | .9941 | .9943 | .9945 | .9946 | .9948 | .9949 | .9951 | .9952 |
| 2.6 | .9953 | .9955 | .9956 | .9957 | .9959 | .9960 | .9961 | .9962 | .9963 | .9964 |
| 2.7 | .9965 | .9966 | .9967 | .9968 | .9969 | .9970 | .9971 | .9972 | .9973 | .9974 |
| 2.8 | .9974 | .9975 | .9976 | .9977 | .9977 | .9978 | .9979 | .9979 | .9980 | .9981 |
| 2.9 | .9981 | .9982 | .9982 | .9983 | .9984 | .9984 | .9985 | .9985 | .9986 | .9986 |

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# Sample Mean

- Assume a population with mean $\mu$ and variance $\sigma^2$.
  - $\mu$ is unknown and is to be estimated, $\sigma$ is known.
  - Neither is known and are to be estimated.
- 'Obvious method': Obtain $n$ independent samples whose values will be $X_1, \ldots X_n$ and calculate the sample mean as an estimator for the mean.

$$\text{Sample Mean} := \hat{\mu}_n := \frac{\sum_{i=1}^{n} X_i}{n}$$

- Clearly, $\hat{\mu}_n$ is a random variable. Characterising this random variable characterises the quality of the estimate.

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# Sample Mean

- The expectation and variance of the sample mean are

$$\mathsf{E}(\hat{\mu}_n) = \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right)$$

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# Sample Mean

- The expectation and variance of the sample mean are

$$\mathsf{E}(\hat{\mu}_n) \quad = \quad \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right) \quad = \quad \frac{\sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i)}{n}$$

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# Sample Mean

- The expectation and variance of the sample mean are

$$\mathsf{E}(\hat{\mu}_n) \quad = \quad \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right) \quad = \quad \frac{\sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i)}{n} \quad = \quad \mu$$

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# Sample Mean

- The expectation and variance of the sample mean are

$$\begin{aligned} \mathsf{E}(\hat{\mu}_n) &= \mathsf{E}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{E}(\mathsf{X}_i)}{n} = \mu \\ \mathsf{VAR}(\hat{\mu}_n) &= \mathsf{VAR}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) \end{aligned}$$

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# Sample Mean

- The expectation and variance of the sample mean are

$$\begin{aligned} \mathsf{E}(\hat{\mu}_n) &= \mathsf{E}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{E}(\mathsf{X}_i)}{n} = \mu \\ \mathsf{VAR}(\hat{\mu}_n) &= \mathsf{VAR}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{VAR}(\mathsf{X}_i)}{n} \end{aligned}$$

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# Sample Mean

- The expectation and variance of the sample mean are

$$\begin{aligned} \mathsf{E}(\hat{\mu}_n) &= \mathsf{E}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{E}(\mathsf{X}_i)}{n} = \mu \\ \mathsf{VAR}(\hat{\mu}_n) &= \mathsf{VAR}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{VAR}(\mathsf{X}_i)}{n} = \frac{\sigma^2}{n} \end{aligned}$$

- What is the distribution of $\hat{\mu}_n$?
- How about 'an approximate distribution' of $\hat{\mu}_n$?

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# Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\mathsf{V} \quad := \quad \sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2$$

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# Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &:= \sum_{i=1}^{n} (X_i - \mu)^2 \\ E(V) &= E\left( \sum_{i=1}^{n} (X_i - \mu)^2 \right) \end{aligned}$$

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# Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
V &:= \sum_{i=1}^{n} (X_i - \mu)^2 \\
E(V) &= E\left( \sum_{i=1}^{n} (X_i - \mu)^2 \right) \\
&= \sum_{i=1}^{n} E\left( (X_i - \mu)^2 \right)
\end{aligned}
$$

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# Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} \mathsf{V} &:= \sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2 \\ \mathsf{E}(\mathsf{V}) &= \mathsf{E}\left( \sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2 \right) \\ &= \sum_{i=1}^{n} \mathsf{E}\left( (\mathsf{X}_i - \mu)^2 \right) \\ &= n\sigma^2 \end{aligned}$$

- Thus if $\mu$ is known then $V/n$ is an unbiased estimator of $\mathsf{VAR}(\mathsf{X}_1)$ .

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^{n} (X_i - \hat{\mu})^2$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\ &= \sum_{i=1}^{n} \left( X_i^2 - 2X_i\hat{\mu} + (\hat{\mu})^2 \right) \end{aligned}$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left( X_i^2 - 2X_i\hat{\mu} + (\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} (X_i^2) - 2\hat{\mu} \sum_{i=1}^{n} X_i + \sum_{i=1}^{n} (\hat{\mu})^2
\end{aligned}
$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

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\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left( X_i^2 - 2X_i\hat{\mu} + (\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \left( X_i^2 \right) - 2\hat{\mu} \sum_{i=1}^{n} X_i + \sum_{i=1}^{n} (\hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left( X_i^2 \right) - 2\hat{\mu}(n\hat{\mu}) + n\,(\hat{\mu})^2
\end{aligned}
$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left( X_i^2 - 2X_i\hat{\mu} + (\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \left( X_i^2 \right) - 2\hat{\mu} \sum_{i=1}^{n} X_i + \sum_{i=1}^{n} (\hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left( X_i^2 \right) - 2\hat{\mu}(n\hat{\mu}) + n\,(\hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left( X_i^2 \right) - n\,(\hat{\mu})^2
\end{aligned}
$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

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\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left( X_i^2 - 2X_i\hat{\mu} + (\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \left( X_i^2 \right) - 2\hat{\mu} \sum_{i=1}^{n} X_i + \sum_{i=1}^{n} (\hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left( X_i^2 \right) - 2\hat{\mu}(n\hat{\mu}) + n\,(\hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left( X_i^2 \right) - n\,(\hat{\mu})^2
\end{aligned}
$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\mathsf{V} = \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\ \sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \end{aligned}$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

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\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \\
E(V) &= E\left( \sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \right)
\end{aligned}
$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

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\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \\
E(V) &= E\left( \sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} E(X_i^2) - nE((\hat{\mu})^2)
\end{aligned}
$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \\
E(V) &= E\left( \sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} E(X_i^2) - nE((\hat{\mu})^2) \\
&= nVAR(X_1) + n(E(X_1))^2 - nVAR(\hat{\mu}) - n(E(\hat{\mu}))^2
\end{aligned}
$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \\
E(V) &= E\left( \sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} E(X_i^2) - nE((\hat{\mu})^2) \\
&= nVAR(X_1) + n(E(X_1))^2 - nVAR(\hat{\mu}) - n(E(\hat{\mu}))^2 \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2
\end{aligned}
$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
\mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 &= \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \\
\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left( \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i^2) - n\mathsf{E}((\hat{\mu})^2) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu}))^2 \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \; =
\end{aligned}
$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
\mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 &= \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \\
\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left( \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i^2) - n\mathsf{E}((\hat{\mu})^2) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu}))^2 \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \;=\; (n-1)\sigma^2.
\end{aligned}
$$

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# Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
\mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 &= \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \\
\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left( \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i^2) - n\mathsf{E}((\hat{\mu})^2) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu}))^2 \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \;=\; (n-1)\sigma^2.
\end{aligned}
$$

- Thus, when $\mu$ is not known, $\frac{V}{n-1}$ is the unbiased estimate of $\mathsf{VAR}(\mathsf{X}_1)$.

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# Summarizing...

- Random independent samples from a population
- Each sample is an independent random variable
- Sample mean is a statistic.
- Expectation of sample mean is the same as the true mean.
- Variance of the sample mean is $\sigma^2/n$
- Since samples are independent, we can invoke LLN and CLT on the Sample Sum.

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# CLT and the Sample Mean

- Let $\mu$ and $\sigma^2$ be the true mean and true variance of a population feature.
- Applying CLT to the sample sum of the $n$ independent sample, $S_n$ we have

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} \rightarrow \mathsf{Z}$$

  where $\mathsf{Z}$ is the unit normal.
- Let $\mathsf{Prob}(\mathsf{Z} > z_\alpha) = \alpha$

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# CLT and the Sample Mean

- Let us now characterize deviation of sample mean from true mean.

$$\text{Prob}\left(-z_\alpha \leq \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})} \leq z_\alpha\right) = 1 - 2\alpha$$

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\text{Prob}\left(|\hat{\mu}_n - \mu| \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha
\end{aligned}$$

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\text{Prob}\left(|\hat{\mu}_n - \mu| \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha \\
\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}} z_\alpha \leq \mu \leq \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha
\end{aligned}$$

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\text{Prob}\left(|\hat{\mu}_n - \mu| \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha \\
\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}} z_\alpha \leq \mu \leq \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha
\end{aligned}$$

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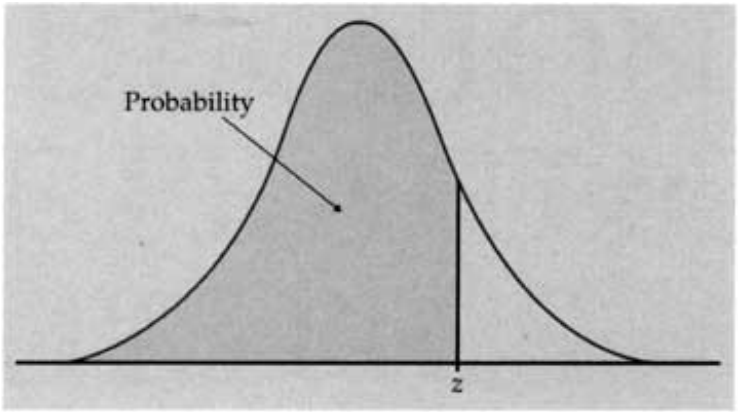



**Table entry for $z$ is the area under the standard normal curve to the left of $z$.**

TABLE A Standard normal probabilities (*continued*)

| $z$ | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| 0.1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| 0.2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| 0.3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| 0.4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| 0.5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| 0.6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| 0.7 | .7580 | .7611 | .7642 | .7673 | .7704 | .7734 | .7764 | .7794 | .7823 | .7852 |
| 0.8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8078 | .8106 | .8133 |
| 0.9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9236 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9484 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9761 | .9767 |
| 2.0 | .9772 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |
| 2.1 | .9821 | .9826 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9854 | .9857 |
| 2.2 | .9861 | .9864 | .9868 | .9871 | .9875 | .9878 | .9881 | .9884 | .9887 | .9890 |
| 2.3 | .9893 | .9896 | .9898 | .9901 | .9904 | .9906 | .9909 | .9911 | .9913 | .9916 |
| 2.4 | .9918 | .9920 | .9922 | .9925 | .9927 | .9929 | .9931 | .9932 | .9934 | .9936 |
| 2.5 | .9938 | .9940 | .9941 | .9943 | .9945 | .9946 | .9948 | .9949 | .9951 | .9952 |
| 2.6 | .9953 | .9955 | .9956 | .9957 | .9959 | .9960 | .9961 | .9962 | .9963 | .9964 |
| 2.7 | .9965 | .9966 | .9967 | .9968 | .9969 | .9970 | .9971 | .9972 | .9973 | .9974 |
| 2.8 | .9974 | .9975 | .9976 | .9977 | .9977 | .9978 | .9979 | .9979 | .9980 | .9981 |
| 2.9 | .9981 | .9982 | .9982 | .9983 | .9984 | .9984 | .9985 | .9985 | .9986 | .9986 |

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# CLT and the Sample Mean

$$\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha \;\leq\; \mu \;\leq\; \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) \quad = \quad 1 - 2\alpha$$

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# CLT and the Sample Mean

$$\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha \;\leq\; \mu \;\leq\; \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) \quad = \quad 1 - 2\alpha$$

- $\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha, \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right)$ is called the $(1-\alpha)$ confidence interval for $\mu$.
- Usually, variance is not known!

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# CLT and the Sample Mean

$$\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha \;\leq\; \mu \;\leq\; \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) \quad = \quad 1 - 2\alpha$$

- $\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha, \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right)$ is called the $(1-\alpha)$ confidence interval for $\mu$.
- Usually, variance is not known! So ...
- Compute

$$\hat{V} = \frac{1}{n-1} \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2$$

and use $\hat{V}$ as the variance, i.e., use $\sigma = \sqrt{\hat{V}}$ in the preceding.

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# The $n\sigma$ rules

- $3\sigma$ and $6\sigma$ rule is something that is commonly used in quality control departments.
- Most manufacturing processes have tolerance; the allowable deviation from the nominal value.
- Production outside of this tolerance band is discarded;
- A $3\sigma$ production process means that deviation from the nominal value are such that $3\sigma$ is within the tolerance ban.
- Lower tolerance requires correspondingly lower $\sigma$ and a tighter control over the process.

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# Some experiments

- Write a program to do the following:
  1. Generate $n$ IID random variables of a specified distribution $F$ whose mean and standard deviation are known.
  2. Compute the sample mean and the limits of the 95% confidence interval.
  3. Check if the true mean is in the 95% confidence interval;
  4. In an outer loop, repeat this experiment $K$ times.
  5. Plot the histogram of the sample mean from the $K$ experiments.
  6. Make a scatter plot of the left and the right limits of the 95% confidence interval.

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# Some experiments

- Let $F$ correspond to Gaussian with $\mu = 0$ and $\sigma = 2$. Let $n = 20$ and $K = 1000$. Repeat for different $\sigma = 3, 5$ and $n = 50, 100, 200, 500, 1000$.
- Now do it for $F$ uniformly distributed in $(0, 1)$, $(-1, 1)$, and $(-2, 2)$.
- And for $F$ being a Bernouilli distribution
- Now pretend $\sigma$ is not known and use $\hat{V}$ as above and see if there is a difference.

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# Sampling from a Normal Population

- If $X_i$ are i.i.d. Gaussian with mean $\mu$ and standard deviation $\sigma$ then

$$\frac{\hat{\mu} - \mu}{\sigma/\sqrt{n}}$$

  is standard normal.
- Define

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \hat{\mu})^2}{n-1}$$

- Can show that $\hat{\mu}$ and $S^2$ are independent random variables.

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# Known Variance

- Since $\hat{\mu}_n$ is Gaussian, we can say

$$\text{Prob}\left(\mu - z_{\delta/2}\frac{\sigma}{\sqrt{n}} \leq \hat{\mu}_n \leq \mu + z_{\delta/2}\frac{\sigma}{\sqrt{n}}\right) \quad = \quad 1 - \frac{\delta}{2} - \frac{\delta}{2}$$

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# Known Variance

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$$\begin{aligned}
\text{Prob}\left(\mu - z_{\delta/2}\frac{\sigma}{\sqrt{n}} \leq \hat{\mu}_n \leq \mu + z_{\delta/2}\frac{\sigma}{\sqrt{n}}\right) &= 1 - \frac{\delta}{2} - \frac{\delta}{2} \\
\text{Prob}\left(\hat{\mu}_n - z_{\delta/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \hat{\mu}_n + z_{\delta/2}\frac{\sigma}{\sqrt{n}}\right) &= 1 - \delta = \gamma
\end{aligned}$$

- In this case we say that $(\hat{\mu}_n - z_{\delta/2}\frac{\sigma}{\sqrt{n}})$ is the $\gamma$ confidence interval for $\mu$.
- If the population is not from a normal distribution, invoke CLT to use the preceding as an approximation.
- Alternatively, use Chebyshev inequality to

$$\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n\delta}} \leq \mu \leq \hat{\mu}_n + \frac{\sigma}{\sqrt{n\delta}}\right) \geq 1 - \delta = \gamma.$$

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# Unknown Variance: Distributions of $\hat{\mu}$ and $S^2$

- $X_i$, $i = 1, \ldots, n$ are iid samples of Gaussian of mean $\mu$ and variance $\sigma^2$.
- Recall sample mean, $\hat{\mu}$, and sample variance, $S^2$, are

$$\begin{aligned} \hat{\mu} &= \frac{\sum_{i=1}^{n} X_i}{n} \\ S^2 &= \frac{\sum_{i=1}^{n} (X_i - \hat{\mu})^2}{n-1} \end{aligned}$$

- Sum of independent Gaussian random variables is also a Gaussian.
- Thus the sample mean $\hat{\mu}$, is Gaussian with mean $\mu$ and variance $\sigma^2/n$.
- Now consider

$$\begin{aligned} \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 &= \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \hat{\mu})^2 + \frac{1}{\sigma^2} n(\hat{\mu} - \mu)^2 \\ &= \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \hat{\mu})^2 + \left(\frac{\sqrt{n}(\hat{\mu} - \mu)}{\sigma}\right)^2 \end{aligned}$$

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# Chi-Square and $t$ Distributions

- If $Z, Z_1, \ldots Z_n$ are independent unit normal random variables then
  - $Z_1^2 + Z_2^2 + \cdots + Z_n^2$ is a $\chi^2$ with $n$-degrees of freedom. It has mean 0 and variance $2n$.
  - The quantity

$$\frac{Z}{\sqrt{(Z_1^2 + Z_2^2 + \cdots + Z_n^2)/n}}$$

    has $t$-distribution. This has mean 0 and variance $n/(n-2)$.
- LHS is $\chi_n^2$. Second term of RHS is $\chi_1^2$. Hence reasonable to guess that first term of RHS is $\chi_{n-1}^2$.

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# Unknown Variance: Distributions of $\hat{\mu}$ and $S^2$

- Can show that $\hat{\mu}$ and $S^2$ are independent with $\hat{\mu}$ being normal and $(n-1)S^2/\sigma^2$ being $\chi^2_{n-1}$.
- This means $\sqrt{n}\frac{\hat{\mu}-\mu}{S}$ is Student $t$ distribution, $t_{n-1}$.
- This helps in obtaining the confidence interval when the variance is unknown.

$$\text{Prob}\left(-t_{\alpha/2,n-1} < \sqrt{n}\frac{\hat{\mu}-\mu}{S} < t_{\alpha/2,n-1}\right) = 1-\alpha$$

- To obtain confidence interval for vairance note that $(n-1)S^2/\sigma^2$ is $\chi^2_{n-1}$. Thus

$$\text{Prob}\left(\chi^2_{\alpha/2,n-1} \leq (n-1)\frac{S^2}{\sigma^2} \leq \chi^2_{\alpha/2,n-1}\right) = 1-\alpha$$

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# Parameter Estimation

- Many times, we have a model for the CDF of pdf of population feature of interest.
- The model CDF will have parameters.
- We will now discuss how to determine the **best estimate** of the paramter(s).
- As before we will obtain $n$ independent sample realizations of the parameter of interest.

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# What is being estimated?




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# What is being estimated?




And what is the method?

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# Likelihood Function

- $x_1, \ldots x_K$ is the set of measurements.
- Assume that the parameter set is $\theta$.
- Question: What is the *likelihood* of observing of $x_1, \ldots x_K$ if the parameter value was $\theta$?
  - Assume the population is Bernoulli with parameter $p$.
  - Assume the population is Poisson with parameter $\lambda$.
  - Assume the population is uniform in the interval $(0, a)$.
  - Assume the population is Gaussian with unit variance and mean $\mu$.
  - Assume the population is exponential with parameter $\mu$.

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# Maximum Likelihood Estimate

- Find the $\theta$ that maximises the likelihood of the observations.

$$\hat{\theta} = \arg\max_{\theta} L(x_1, \ldots, x_K|\theta)$$

- Can also maximise the Log Likelihood function.
- Likelihood and log likelihood functions and MLEs
- Bernouilli

$$\begin{aligned} L(\underline{x}; p) &= \prod_{i=1}^{K} p^{x_i}(1-p)^{1-x_i} = p^n(1-p)^{K-n} \\ \hat{p} &= \frac{\sum_{i=1}^{K} x_i}{K} \end{aligned}$$

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